

Team Contribution Report

Deep Learning Final Project · AI Academy · Cohort I 2026
Project: Azerbaijani Banknote Vision (AZN Vision) | Repository: v1.0-final

1. Team Contribution Summary Matrix

Member Name	Code (%)	Report Sections	Experiments	Slides (%)	Other Responsibilities
Avaz	30%	Abstract, Architecture, TinyML, Discussion	Exp 1 (Battle), TinyML Edge	25%	Lead Architect, CI/CD, Git
Gulnar	25%	Dataset Methodology, Augmentations	Exp 2 (Augmentation), Exp 4 (Resolution)	25%	Data Collection, Quality Audit
Hasan	25%	Related Work, Color Shortcuts, Limits	Exp 3 (Color Shortcuts), Exp 5 (Quantization)	25%	Photometric Profiling, Asymmetric Loss
Kazim	20%	Explainability (XAI), Deployment	Exp 6 (XAI Attention), Mobile App	25%	EigenCAM, Telemetry Dashboard
Total	100%	Full IEEE Paper	6 Core Studies + TinyML	100%	Balanced Team Allocation

2. Individual Member Contributions

Avaz (Lead Architect & Systems Engineer): Designed and maintained the core repository structure, zero-leakage stratification pipeline (splitter.py), and DINOv2 feature clustering. Implemented and tuned TinyML YOLO-FastestV2 edge pipeline with CIoU loss, asymmetric focal classification, and INT8 calibration. Integrated Sseed Studio XIAO ESP32-S3 embedded C++ firmware and synchronized C++ model headers. Developed the local GPU inference bridge and unified reproducibility harness (run_all.py).

Gulnar (Data Lead & Augmentation Specialist): Coordinated the physical capture and labelling of 2,593 raw banknote images across 93 real-world scenes. Implemented and evaluated Experiment 2 (Data Augmentation Ablation: Raw vs. Geometric vs. Photometric vs. Full Composite). Conducted dataset integrity audits, boundary box coordinate verification, and perceptual duplicate pruning. Contributed to Experiment 4 (Resolution Scaling Dynamics: 320 vs. 640 vs. 1280).

Hasan (Photometric & Robustness Specialist): Conducted deep photometric profiling and color channel variance analysis across Azerbaijani currency denominations. Implemented and analyzed Experiment 3 (Color Space & Shortcut Learning: RGB vs. HSV vs. Grayscale), proving the network relies on authentic geometric guilloche engraving rather than color hue. Implemented the asymmetric cost-sensitive penalty vector penalizing high-denomination false positives. Contributed to post-training quantization and model pruning benchmarks in Experiment 5.

Kazim (Explainability & Deployment Specialist): Implemented EigenCAM attention visualization for Experiment 6, proving that C2PSA attention heads align with numismatic security motifs. Contributed to the real-time telemetry dashboard (dashboard/) and dynamic ETA estimation engine. Designed and verified UI components for the React Native mobile companion application (mobile/). Conducted live mobile testing and validation on physical Android and iOS devices.

3. Team Agreement & Sign-Off

All four team members have reviewed, agreed to, and verified the contributions documented in this report. The work presented in the repository, paper, and slide deck reflects genuine collaborative effort.